Detour Ahead: When Correlation Misleads

Checkpoint 1 Report

1. Meta Information

Project Title	Detour Ahead: When Correlation Misleads
Group Number	"029"
Mentor	Aakash Gupta
Members (Roll No.)	Gururatna (25b0906), Nigam (25b0958), Tripathi (25b0965), Kirti (25b1059)
GitHub Repository	`https://github.com/kirtiydv/dai_project/`
Tracking Spreadsheet	`https://drive.google.com/drive/folders/1WStrpDyWQztSkLnUoQ0yJejXTfocL-Vg` (our mentor has edit access)

2. Problem Statement

Correlation between two variables does not necessarily imply causation. Strong statistical associations may arise from shared trends, seasonality, or confounding factors, even when no meaningful causal relationship exists.

Our project investigates this phenomenon using three variables in India:

- **A – Road Infrastructure Expenditure:** Monthly government spending on road development/maintenance.
- **B – Urban Traffic Speed:** Average traffic speeds in Mumbai and Bengaluru.
- **C – Demat Accounts:** Monthly number of registered demat investor accounts.

Research Question:

To what extent do correlations among road expenditure, urban traffic speed, and demat account growth reflect meaningful associations, rather than shared trends, seasonality, or other temporal effects?

We aim to analyze these relationships using Pearson and Spearman correlations, detrending, differencing, seasonal adjustment, and lagged correlation analysis. The objective is to identify potentially spurious correlations and demonstrate why statistical association alone cannot establish causality

3. How we got the raw data for B and C

B (traffic speed). We used the TomTom Traffic Index pages for Mumbai and Bengaluru (2025 edition, accessed on 20 Oct 2026). From each page we copied, exactly as shown, the *monthly congestion level* chart (24 values per city, Jan to Dec for 2024 and 2025) and the yearly figures: average travel time for 10 km, rush hour speed, share of highway trips and the worst day of the year. *Getting more:* archived snapshots of older TomTom editions (2022 and 2023) and a few more Indian cities.

C (demat accounts). We used the annexure tables (Excel) of the SEBI Monthly Bulletin, specifically the "Progress Report of NSDL & CDSL" table and its "Number of Investor Accounts" row (in lakh). Each issue reports three months (the latest month, the month before it, and the same month a year earlier), so we downloaded 14 alternate issues from Feb 2024 to Apr 2026. That gives us 34 months, from Jan 2023 to Mar 2026, with no gaps from Nov 2023 onwards. *Getting more:* we wrote a download script to pull issues back to 2019 (around 90 months), and we will cross-check them against CDSL and NSDL's own monthly statistics.

4. Cleaning done so far for B and C

- **B:** TomTom's chart lists the two years side by side for every month, so we first sorted the 24 values into the right month and year. We confirmed this ordering by checking that the 12-month averages match TomTom's published yearly congestion (within 0.2 points for both cities). We then converted congestion into average speed (see Section 4) and stored the raw page values separately from the converted ones.
- **C:** The account counts are not in the same place every year. The table number moves (Table 61 in 2024, Table 63 in 2026) and the row name changes slightly ("Investor" vs "Investors" Accounts), so our script searches for the row in each file instead of using a fixed position. We added NSDL and CDSL together for each month. One 2026 header has a year typo, so months are worked out from the latest month rather than read from the header. When a month appears in two issues, we keep the value from the issue where it is the latest month.

5. Plan for refining the data

- **B:** Convert congestion c into average speed using $v = 600/\big(T_{\text{ff}}(1+c)\big)$ km/h, where the free-flow time T_{ff} for 10 km comes from TomTom's yearly travel time and congestion.
- **C:** Keep a log of months where two issues disagree (SEBI sometimes revises figures), and convert the totals into crore and million for readability.
- **All three:** Line up A, B and C on one monthly timeline and compute Pearson and Spearman correlations on (i) the raw series, (ii) the series with the trend removed or as month-to-month changes, and (iii) the seasonally adjusted series. We will also try lagged versions, with A leading B by 3 to 6 months, and report 95% confidence intervals and permutation test p-values rather than just r.

6. Plan for A

The script for A is getting ready. At the city level we are using BBMP's (Bengaluru) actual monthly payments from its accounts portal, shared through OpenCity. The script reads the payment date from each record, tags each work as road related or not, and adds everything up month by month. This should give us usable months from Apr 2024 to Aug 2025, which overlap nicely with B. At the national level we plan to use MoRTH's monthly expenditure (CGA accounts) to build a much longer A and C series, around 10 years. Mumbai's yearly BMC budgets will only be used as background.

7. Data checks and known limitations

- **Checks:** TomTom's monthly figures reproduce its yearly congestion within 0.2 points, and our converted 2025 speed for Mumbai (20.8 km/h) matches TomTom's own number. The SEBI totals match figures reported elsewhere (19.24 crore in Mar 2025 and 22.45 crore in Mar 2026).
- **Small sample:** A and B only overlap for about 17 months, so a correlation needs to be above roughly 0.48 to be significant at 5%. We will treat A and B as an illustration rather than a final answer.
- **Trends and seasonality:** demat accounts go up every single month, and road spending spikes every March. Both of these can create correlations that are not real, which is exactly the effect this project is about.
- **Data caveats:** the monthly speed is something we calculate, not a figure TomTom publishes. Road spending will be a range, because many works are listed as "roads and drains". BBMP data after Aug 2025 looks incomplete, and SEBI's Aug 2024 NSDL value is the same as Jul 2024.

8. Team contribution

Member	Work done so far
Gururatna	Analysis of trends in datasets of B and C, Managing submissions, Feasibility check
Nigam	Finding datasets for B and C, Scraping site for Data of A, Datasets Trend Analysis
Tripathi	Finding datasets for B and C, Cleansing datasets, Scraping Site for data of A
Kirti	Dataset cleansing, Repository management, Finding Sources for data of A

9. GitHub repository (data part)

Link: `https://github.com/kirtiydv/dai_project/`
Rule we follow: files in `raw/` are never edited. Anything cleaned or converted goes into `processed/`.

Folder	What is inside
`data/raw/A_bbmp_payments/`	BBMP work order and payment CSVs (FY 2024-25 and 2025-26) with SOURCES.txt
`data/raw/B_traffic_TomTom/`	tomtom_mumbai_raw.xlsx and tomtom_bengaluru_raw.xlsx with SOURCES.txt
`data/raw/C_demat_SEBI/`	14 SEBI bulletin annexure files (3 months each) with SOURCES.txt
`data/processed/`	Monthly A, B and C series and merged_monthly.csv, all generated by the scripts in src/